Lesson notes · for you

Messy names, wrong counts

Data engineering · Level 2 · Lesson 2

Time	Materials
15 min	Your phone with Darsi, for the Python steps; Paper and a pen

Learning goals

- Spot spaces, capitals and spellings that break counts.
- Clean step by step: tidy, then match a list.
- Clean text in Python with strip() and lower().

Activity: Clean the poll

1. Vote (3 min): Everyone writes their favourite sport on a slip of paper, with no rules about spelling.
2. Count (4 min): Count the answers exactly as written, the way a computer would.
3. Clean (5 min): Agree one spelling for each sport, and sort the slips into those groups.
4. Recount (3 min): Count again and compare. Did the winner change?

Encourage different spellings on purpose, so there's plenty to clean.

Discussion

- Why does a computer count "Zoo" and "zoo" separately?
- Which cleaning step found the most differences?
- How would a drop-down list help?

Check yourself

- Write one answer three ways a computer would count as three, and its clean form.

Try it this week

Look at a sign-up sheet or a class poll where people typed their answers. Count how many ways one answer was written. Then write the clean list.

Worksheet

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Name _____ Class _____ Date _____

1 **Which of these stops messy answers before they start?**

◯ A bigger text box ◯ Asking people to type carefully ◯ A drop-down list to choose from

2 **Which of these will a computer count as different from "zoo"? Choose all that apply.**

☐ "ZOO" ☐ "zoo " (with a space at the end) ☐ "Zoo" ☐ "zoo"

3 **Fill the gaps so every vote is tidied before it's counted.**

```
vote = " Zoo "
clean = vote._____().  _____()
print(clean)
```

Word bank: `upper` `strip` `count` `lower`

4 **Put the lines in order (and indent them): the program tidies each vote and counts the votes for the zoo.**

Number the lines in order (1, 2, 3…), and draw one arrow → in front of a line for each indent it needs. Cross out any line that doesn't belong.

☐ `print(counts["zoo"])`

☐ `clean = vote.strip().lower()`

☐ `for vote in votes:`

☐ `clean = vote`

☐ `counts = {}`

☐ `counts[clean] = counts.get(clean, 0) + 1`

5 **The class voted for a trip by typing. Fix the program so it tidies every vote before counting: it should print the votes for zoo, then for museum.**

```
votes = ["Zoo", "zoo ", "museum", " Zoo", "farm", "ZOO", "Museum", "zoo"]
print(votes.count("zoo"))
print(votes.count("museum"))
```

Write your program:

✎ **Where have you seen the same answer typed in different ways: in a game, a form or a class poll?**

Check your answers

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1 **A drop-down list to choose from**

When people choose from a list instead of typing, every answer is written the same way, so there's nothing to clean.

2 **"Zoo"; "zoo " (with a space at the end); "ZOO"**

Only an exact copy matches. A capital, a space at the end or a different letter makes a new answer to the computer.

3

```
vote = " Zoo "
clean = vote.strip().lower()
print(clean)
```

strip() removes the spaces at both ends, then lower() makes the letters small: the program prints zoo.

4 **Top to bottom: 5, 3, 2, ✗, 1, 4**

```
counts = {}
for vote in votes:
    clean = vote.strip().lower()
    counts[clean] = counts.get(clean, 0) + 1
print(counts["zoo"])
```

Start with an empty dictionary. For each vote, tidy it first, then add one to that answer's count.

5

```
votes = ["Zoo", "zoo ", "museum", " Zoo", "farm", "ZOO", "Museum", "zoo"]
clean = []
for vote in votes:
    clean.append(vote.strip().lower())
print(clean.count("zoo"))
print(clean.count("museum"))
```

As typed, count finds only the exact "zoo": 1.